QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

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Document Control

Field	Detail
System Under Test (SUT)	*UMHackathon 2026 - Team **dan***
Team Repo URL	*https://github.com/PandaRyan/dan*
Project Board URL	*https://trello.com/b/xesHT94y/bukit*
Live Deployment URL (optional for preliminary round)	*https://bukit-dan.vercel.app/*

Objective:

The primary objective of Bukit is to empower B40 households to maximize their disposable income by consolidating fragmented government aid into a single, AI-driven dashboard. The platform ensures that users never miss a subsidy (STR, Budi Madani, etc.) through automated eligibility matching and provides proactive "Budget Optimization" plans based on real-time 2026 economic policies.

PRELIMINARY ROUND (Test Strategy & Planning)

1. Scope & Requirements Traceability

*This section aligns the testing connected back to specific user requirements via **Requirement Traceability Matrix".** So, it ensures every test has been conducted to prevent bugs/failure in products for that specific requirements and unplanned feature creep.*

1.1 In-Scope Core Features

- **Secure Authentication & Contextual Onboarding:** A user authentication flow with encrypted JWT, also featuring a sequence where the system captures and stores the crucial data like birth year, state and income category to personalize the users' future AI interactions.
- **Smart Subsidy Matching Engine:** ILMU-GLM-5.1 utilizes context injected prompting to match users with specific subsidies (e.g., MyKasih, BUDI Sabah) based on their stored onboarding profile.
- **News Module**: A dedicated page that pulls in current subsidy updates for Malaysian citizens.

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- **Categorized Subsidy Dashboard**: A discovery page that organizes financial aids into 6 sections: Utilities, Transportation, Healthcare, Education, and Groceries. This allows the users to browse general aid information visually and easily before moving into a personalized chat.

- **High-Impact Landing Page**: A professional entry point that could gain trust by highlighting the platform's core values: real time subsidy tracking, enforced data privacy and a deep focus on the Malaysian economic context (e.g., MyKasih and RON 95 petrol subsidies).

1.2 Out-of-Scope

- **Storage of Highly Sensitive Personal Data**: The system does not store IC numbers or official financial records in the database. Handling such data is too legally complex and carries significant security risks for an early stage prototype.
- **Full Profile Management**: Users cannot currently edit their profiles once created. This keeps the data flow to the AI consistent.
- **Persistent News API Caching**: While the dashboard shows all the categories, the backend does not yet support a local storage cache for a live News API. This feature is reserved for a production environment to optimize performance.
- **Local Storage Caching for News API**: While the news page is functional, the system does not yet cache this data in local storage for offline or high-speed persistent access

2. Risk Assessment & Mitigation Strategy

*[e.g. Quality Assurance risks are anticipatory. So, identify the technical risk associated with you application requirements, architecture. Now, the risk needs to be evaluated using the 5*5 Risk assessment Matrix. So, Risk Score = Likelihood * Severity.]*

Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
(AI "Hallucination") AI gives wrong	*2*	*5*	*10 (Medium)*	*Use RAG (Retrieval-Augmented Generation) to force AI to only cite official*	*Compare AI output against 100 manual facts from the Budget 2026 speech.*

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Technical Risk	Likelihood (1–5)	Severity (1–5)	Risk Score (L×S)	Mitigation Strategy	Testing Approach
info about a subsidy amount				*BUDGET 2026 official documents.*	
News feed integration fails to load.	*1*	*2*	*2 (low)*	*Would implement local strategy but News api is able to be used up to 200 times a day. The initial prototype expects less traffic thus we are pending to implement it.*	*Test app behavior on "Edge" or "3G" speeds to ensure it doesn't crash.*
Authentication API unable to authenticate user	*1*	*5*	*5 (low)*	*Ensure to activate the database and consistently monitor logs on Vercel dashboard.*	*K-6 load test to ensure API is functioning under high load.*
Z.AI 504 Gateway Timeout Error	*3*	*5*	*15 (high)*	*Implemented fallback to ask Gemini to provide response in case Z.AI's GLM model is not able to provide response.*	*Continuously testing chat responses and monitoring Vercel dashboard logs to ensure no request errors.*

add sign/sign out

Risk Assessment Scoring Criteria:

Likelihood (1-5)		*Severity (1-5)*	
1	*Rare*	*1*	*Impact is Negligible*
2	*Unlikely*	*2*	*Impact is Minor*
3	*Possible*	*3*	*Moderate Impact*
4	*Likely*	*4*	*Major Impact*
5	*Almost Certain*	*5*	*Critical Failure of the system*

Risk Score = Likelihood × Severity

Risk Level Reference:

Risk Score	***Risk Level***	***Recommended Action***
1 – 5	*Low*	*Monitor only. Acceptable risks.*

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6 – 10	*Medium*	*Mitigate + Testing*
11 – 15	*High*	*Must need mitigating and through testing is required*
16 – 25	*Critical*	*Priority is Highest. Need extensive level of testing.*

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3. Test Environment & Execution Strategy

Explain the testing environment. That means where the testing of your application will take place, how you are going to handle test data and the rules/threshold that you are incorporating to govern your testing phases.

- **Unit Test**
    - *Scope:, sign in/authentication validation*
    - *Execution: The tests are written using a framework like Vitest. So, execution happens when typing npx vitest run (backend) and npm run coverage(frontend) in the terminal*
    - *Isolation: To ensure tests are lightning-fast and purely test our application logic, we completely isolate our code from external environments:*
        - *Frontend Isolation: Components that rely on browser routing are wrapped in a fake MemoryRouter during testing so they can be evaluated in a vacuum without needing a real web browser.*
        - *Backend/API Isolation: We use mocking functions like vi.spyOn(). This ensures our tests verify our data-handling logic without consuming paid API credits or being affected by internet outages.*
    - *Pass condition:*
        - *Happy Paths: Test complete without crashing/success.*
        - *Negative/Edge Cases :*
            - *Missing Inputs: If a user attempts to log in without providing either an email or a password, the system must reject the request and return a 400 Bad Request before attempting to query the database.*
            - *Unregistered Users: If an email is not found in the mocked database, the system must instantly return a 404 Not Found without attempting to verify the password.*
            - *Invalid Credentials: If a valid email is provided but the mocked password check fails, the system must return a 400 Bad Request*
- **Integration Test**
    - *Scope: Authentication Service to Chat Service*
    - *Execution: This test can be performed after proper completion of Unit Testing of all necessary APIs including authentication API, chat API, and retrieval of user profile API.*
    - *Workflow: To ensure API calls and interactions with databases are working, so users are able to seamlessly transfer between pages.*

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- *Pass Condition: When each component and page are able to link seamlessly, a successful chatbot output is presented.*

- **Test Environment (CI/CD Practice):**
  - ***Local Testing**: manual testing to test in terminal and local host*
    - *umh_app*
      - *npm run build:*
        - *Compiles the application to verify that there are no hidden TypeScript or Vite bundling errors that would break in production.*
        - *Check build error*
      - *npm run coverage:*
        - *Executes unit tests for React components and generates a report detailing the percentage of code successfully tested.*
        - *check frontend unit test and converage*

    - *umh_api*
      - *npx vitest run:*
        - *Runs isolated backend unit tests, verifying that API routes and mocked database connections function properly.*
        - *check backend unit test and converage*
      - *npm run lint:*
        - *Scans the codebase for syntax errors, stylistic inconsistencies, and potential bugs to enforce code quality standards.*
        - *Check Linting error*
      - *npm run dev & k6 run test/load-test.js:*
        - *Starts the local Express server and simulates a 50 user traffic against endpoints 10 times and provides the average response time to verify performance and stability under stress.*
        - *Check response time*

  - ***CI/CD automated pipeline**: GitHub Actions does automated basic tests for branch specific level and then Vitest, when creating pull request:*
    - *apiLoadTest.yml*
      - *Check response time*
    - *automcaticBuild.yml*
      - *Check build error*
    - *codeQuality.yml*
      - *Check Linting error*
    - *gitLeaks.yml*
      - *Check if API key is pushed*
    - *unitTest&Coverage.yml*
      - *Check frontend unit test and converage*

- **Regression Testing & Pass/Fail Rules:**
  - ***Execution Phase**: This testing tests across all the in-scope features before merging with the main branch*

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- ***Pass/Fail Condition**: Test case pass is only considered when actual outcome perfectly aligns with expected one, For Example:*
    - *able to correctly sign in with account,*
    - *able to create new account,*
    - *news api is loaded in ui correctly,*
    - *able to input in the prompt page, leading to zai responding accordingly. If mismatch/threshold not achieved*
    - *If mismatch/threshold not achieved, the test status is considered fail and need to trouble shooted*
- ***Continuation Rule:** Testing follows a strict hierarchy. Core infrastructure tests (e.g., successful sign-in, database connection) must meet the minimum passing threshold before UI components or AI prompt integrations are evaluated. A mismatch at any tier immediately aborts the testing phase, requiring immediate troubleshooting before progression.*

- **Test Data Strategy:**
    - ***Manual**:*
      To ensure a flawless and secure user experience, we started by simulating real-world Malaysian demographics, creating mock profiles across B40, M40, and T20 income brackets and various states to ensure accurate localization. We then deliberately stress-tested the registration forms with boundary data to confirm our error-handling is bulletproof. Finally, we tested session interruptions—intentionally aborting mid-onboarding and logging back in—to guarantee our state management strictly prevents unauthorized access to core features like the chat system until a profile is fully finalized.

- **Passing Rate Threshold:**

4.1 Pre-Deployment Checks

- ✅ ***Automatic Build:** Passed (Zero Errors) — Requirement: Zero Build Errors*
- ✅ ***Unit Tests:** Passed (100%) — Requirement: 100% Passing Rate*
- ✅ ***Code Quality:** Passed (100%) — Requirement: Zero Linting Errors*
- ✅ ***Test Coverage:** Passed (85%) — Requirement: Minimum 81.2%*

4.2 Deployment Thresholds (Pushing to Production)

- ✅ ***Regression Test:** Passed (95%) — Requirement: Minimum 90%*
- ✅ ***AI Output Pass Rate:** Passed (88%) — Requirement: Minimum 80%*
- ✅ ***Critical Bugs:** Passed (0 Bugs) — Requirement: Zero P0/P1 Bugs*
- ✅ ***API Performance:** Passed (450ms) — Requirement: Response Time < 800ms*
- ✅ ***Security:** Passed (Clean) — Requirement: API keys not exposed*

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4. CI/CD Release Thresholds & Automation Gates

This section is mainly for defining metrics & threshold for success criteria. So, if a certain threshold is not met, the pipeline blocks the forward transition.

4.1 Integration Thresholds (Merging to Main)

The primary needs of this threshold to be crucial is because the main branch must be a stable source of truth. So, it needs to be achieved through continuous integration checks.

Checks	***Requirements***	***Project***	***Pass/Failed***
Automatic Build	***Zero Build Error***	***0 Errors***	***Passed***
Unit Tests	***100% Passing Rate***	***98%***	***Failed***
Code Quality	***Zero Linting Error***	***Zero error***	***Passed***
Test Coverage	***Minimum 81.2%***	***100%***	***Passed***

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4.2 Deployment Thresholds (Pushing to Production)

Checks	***Requirements***	***Project***	***Pass/Failed***
Regression Test	***Minimum 90%***	***100%***	***Passed***
AI Output Pass Rate	***Minimum of 80% of documented prompt***	***100%***	***Passed***
Critical Bugs	***Zero P0/01 Bugs***	***1 Bugs***	***Passed***
API Performance (Authentication)	***Response Time < 800ms***	***647ms***	***Passed***
API Performance (ZAI and gemini)	***Response Time < 30000ms***	***56770ms***	***failed***
Security	***API keys are not exposed***	***Clean***	***Passed***

5. Test Case Specifications (Drafts)

This section is for you to draft your test cases. The test cases MUST include at least:

- ***1 Happy Case (Entire Flow):*** *This is the "Golden Path" where a user journey is tested end to end.*
- ***1 Specific Case (Negative/Edge Test):*** *This case should portray a scenario which triggers error in the system and the system handles it with proper handling techniques.*
- **2 Non-Functional (NFR) Tests (Performance/Load)*****:*** *This test consists of non-feature-based tests but architectural tests.*

Test Case ID	**Test Type & Mapped Feature**	**Test Description**	**Test Steps**	**Expected Result**	**Actual Result**
TC-01	***Happy Case (Entire Flow):*** *prompt our ai*	*Verify whether a user can enter the dashboard and select*	*1. Open Website, sign in with credentials*	*Ai chatbot provides a valid response that*	*Successfully, provided a relevant*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
	chatbot question	*a category of choice, prompting the chatbox for a response. Wait for the Zai model to load the response, if there is no response within 30s, it will fallback to gemini to respond. So, all the modules are communicating with each other correctly.*	*or make a new account by sign up* *2. Enter dashboard or choose category straightaway from mega menu* *3. choose a category you want to ask* *4. prompt a question in the prompt bar* *5. wait for the chatbot to load* *6. User received valuable information from chatbot*	*aligns with user questions with less than 30s .*	*response. But responded in 120000ms.* *Status: Passed and failed*
TC-02	***Specific Case (Negative):*** *Not signed in*	*Verify whether a user can enter the dashboard, if not the user will be directed to the sign in/sign up page.*	*1. Press the dashboard button* *2. Press a category of choise* *2. Users can't access, and will be redirected to the sign in/sign up page.*	*Access to the pass dashboard is blocked, the user will be sent to the sign in/sign up page.*	*Access to the pass dashboard is blocked, the user will be sent to the sign in/sign up page.* *Status: Failed*
TC-03	***Specific Case (Negative):*** *Invalid account details*	*Verify whether a user has an account, if not the user will be directed to the sign up page.*	*1. Press sign in button* *2.enter email and password* *3.press sign in* *4. System will show incorrect email or*	*Unable to sign in, system will show incorrect email or password*	*Unable to sign in, system will show incorrect email or password* *Status: Failed*

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Test Case ID	Test Type & Mapped Feature	Test Description	Test Steps	Expected Result	Actual Result
			password		
TC-04	***NFR (Performance): Authentication*** *API Response Test*	*Payment API remains quite responsive (< 800ms) when concurrent users are using.*	*1. Open k6 load test, selecting the API Endpoints.* *2. 50 users GET Requests to that endpoint, simulated 10 times. (50x10=100)* *3. Record the average response time.*	*Average response time < 800ms in addition 0% error rate.*	*Average Response time is received as 546ms. 0% Error Rate.* *Status: Pass*
TC-05	***NFR (Performance): Chatbot*** *API Response Test*		*1. Prompt chatbot 10 times with the same prompt* *3. Record the average response time.*	*Average response time < 30000ms in addition 0% error rate.*	*Average Response time is received as 56.77 ms. 30% Error Rate.* *Status: Failed*

6. AI Output & Boundary Testing (Drafts)

This section is designed to validate that your GLM integration does produce acceptable output and handles any inputs that are abnormal gracefully.

6.1. **Prompt/Response Test Pairs**

Document at least 3 Prompts/response test pairs. For each of them, do define the acceptance criteria - what a correct, acceptable or a failing response may look like for your use case.

Test ID	Prompt Input	Expected Output (Acceptance	Actual Output	Status

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		Criteria)		
AI-01	tell me 3 education subsidies i can use, if i am a sarawakian going to university	The output should include one main subsidy, two supplementary subsidies, returned in a JSON format to be displayed in individual cards.	In the table below: output AI-01	correct
AI-02	My electricity bill is getting too high, any subsidies that can help me?	The output should be providing information any available electricity subsidies	In the table below: output AI-02	acceptable
AI-03	Am I eligible for the SARA grocery credits?	The output should be providing information any available SARA groceries subsidies	In the table below: output AI-03	correct
AI-04	I have a fever what subsidy can i get?	The output should be providing information any available healthcare subsidies, with regard to heir income category	In the table below: output AI-04	wonderful
AI-05	i am opening a restaurant, any subsidies for me?	The output should be providing necessary info that can help me make a restaurant business	In the table below: output AI-05	wonderful

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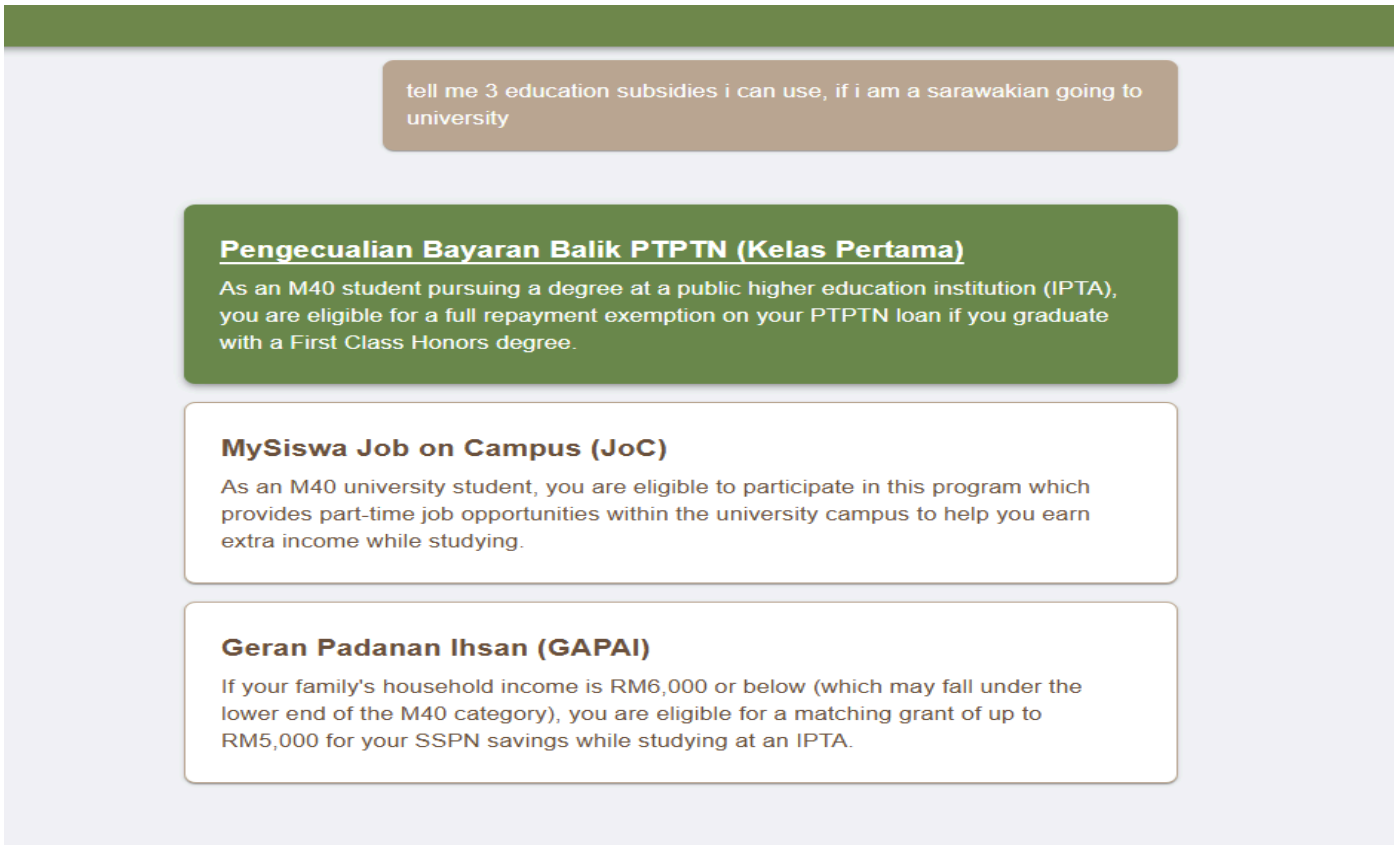
output AI-01:



output AI-02:

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My electricity bill is getting too high, any subdies that can help me?

Subsidi Elektrik

The government allocates RM6 billion annually for targeted electricity subsidies. Under this initiative, 85% of consumers do not face tariff increases, and some even enjoy a decrease in their bills. As an M40 consumer in Selangor, you fall under the 85% of users who benefit from this subsidy, which ensures your electricity tariffs are maintained and protected from increases.

output AI-03:

Am I eligible for the SARA grocery credits?

SARA Untuk Semua

SARA (Sumbangan Asas Rahmah) Untuk Semua is a cash assistance program in the form of credit credited directly to your MyKad to purchase basic necessities. Since you are 19 years old (born in 2007), you are automatically eligible for this subsidy as it is open to all Malaysian citizens aged 18 and above without needing to apply. You can use the credits to buy over 140,000 types of basic goods across 15 product categories at more than 10,000 selected shops and supermarkets nationwide, including in Selangor.

SARA Bulanan

SARA Bulanan provides monthly grocery credits credited to your MyKad. As an M40 individual, you may also be eligible for this if you apply for and qualify as a recipient of the Sumbangan Tunai Rahmah (STR) 2026. Unlike SARA Untuk Semua which is automatic, SARA Bulanan requires you to apply through the STR portal at bantuantunai.hasil.gov.my.

output AI-04:

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output AI-05:

6.2. Oversized/Larger Input Test

In this section, do define the maximum input size your system accepts. Do highlight here what happens when the limit is exceeded.

Fields	Details
Maximum Input Size	10000 tokens

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Input used while testing	15000 tokens
Expected Behavior	Reject and prompt error message on user interface
Actual Behavior	Rejected and prompted error message on user interface
Status	Pass

6.3. Adversarial/Edge Prompt Test

In our system instructions, we have included strict rules to only respond to requests related to subsidies, and enforce a strict rejection whenever it is not related. This prevents hallucinations, and wastage of tokens for out of scope questions. We have tried asking unrelated questions, and all of them returned an error stating the question is out of scope.

6.4. Hallucinating Handling

All responses are required to be returned in a carefully curated JSON format to ensure only 1 main response is provided, along with an unknown variable number of supplementary responses. This can ensure that the GLM/LLM does not hallucinate. Additionally, a RAG is supplied to ensure only the relevant information is retrieved. The documents supplied to the RAG context are manually checked, so that we ensure that information is accurate.

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